

Basic Data Exploration and Cleaning using Pandas

Submitted By

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Objective

The objective of this project is to learn Python basics and perform basic data exploration and cleaning using the Pandas library.

In this project, an ecommerce dataset containing product information such as product names, ratings, prices, discounts, seller information, and categories is used for data analysis and preprocessing.

The following tasks are performed in this notebook:

- Loading the CSV dataset into a Pandas DataFrame
- Exploring the dataset structure and contents
- Identifying and handling missing values
- Performing filtering and column selection operations
- Removing duplicate records
- Creating a derived column (`total_amount`)
- Saving the cleaned dataset as a new CSV file

The dataset used in this project is the Combined Ecommerce Dataset containing around 1000 product records from multiple ecommerce categories.

Importing Required Libraries

In this section, the necessary Python libraries required for data analysis and data cleaning are imported.

- Pandas is used for data manipulation and analysis.
- NumPy is used for numerical operations.

```
import pandas as pd
import numpy as np
```

Loading & Exploring the Dataset

In this section, the ecommerce dataset is loaded into a Pandas DataFrame using the `read_csv()` function.

As well as we will understand the structure of the dataset by checking:

- First and last rows
- Shape of dataset
- Column names
- Data types
- Statistical summary

```
df = pd.read_csv("Combined_dataset.csv")
```

Display the first few rows of the DataFrame to understand its structure

```
df.head()
```

	product_id		title
product_description \			
0	8376765	Lino Perros	Women Navy Blue Solid Backpack
1	9136281	Tommy Hilfiger	Unisex Navy Blue Striped Backpack
2	17633752	Lavie	Aries Women Pink Mini Backpack
3	1376949	F Gear	Unisex Navy & Grey Printed Burner Backpack
4	13939916	MYTRIDENT	Men Blue Solid Bath Robe

	rating	ratings_count	initial_price	discount	final_price
currency \					
0	3.8	15	3995	58.0	"₹3,995.00"
INR					
1	4.5	67	2899	35.0	"₹2,899.00"
INR					
2	4.4	226	2999	65.0	"₹2,999.00"
INR					
3	4.4	1052	1675	52.0	"₹1,675.00"
INR					
4	4.7	12	2899	17.0	"₹2,899.00"
INR					

	images	...	\
0	http://assets.myntassets.com/assets/images/837...	...	
1	http://assets.myntassets.com/assets/images/913...	...	
2	http://assets.myntassets.com/assets/images/176...	...	
3	http://assets.myntassets.com/assets/images/137...	...	

```

4 http://assets.myntassets.com/assets/images/pro... ...
                                amount_of_stars
what_customers_said \
0 {"1_star":2,"2_stars":0,"3_stars":3,"4_stars":...
NaN
1 {"1_star":3,"2_stars":4,"3_stars":2,"4_stars":...
NaN
2 {"1_star":9,"2_stars":5,"3_stars":10,"4_stars"...
NaN
3 {"1_star":33,"2_stars":21,"3_stars":69,"4_star...
NaN
4 {"1_star":0,"2_stars":0,"3_stars":0,"4_stars":...
NaN

seller_name                                sizes \
0      NaN                                [{"size":"Onesize"}]
1      NaN                                [{"size":"Onesize"}]
2      NaN                                [{"size":"S"}]
3      NaN                                [{"size":"Onesize"}]
4      NaN [{"size":"M","value":"45.0in","value_name":"Fr...

                                videos
seller_information \
0                                ["rw-8376765_cae700"]
NaN
1                                ["rw-9136281_cae700"]
NaN
2 ["https://videos.myntassets.com/assets/videos/...
NaN
3 ["https://videos.myntassets.com/assets/videos/...
NaN
4                                NaN
NaN

                                variations best_offer \
0                                [{}]                {}
1                                [{},{]}            {}
2                                [{},{},{},{},{},{]}  {}
3                                NaN                  {}
4 [{},{},{},{},{},{},{},{},{},{},{},{},{},{},{},{]} {}

                                more_offers  category
0 [{"offer_name":"10% Instant Discount on Citi C... backpacks
1 [{"offer_name":"10% Instant Discount on Citi C... backpacks
2 [{"offer_name":"10% Instant Discount on Citi C... backpacks
3 [{"offer_name":"10% Instant Discount on Citi C... backpacks
4 [{"offer_name":"10% Instant Discount on IndusI... bath-robe

[5 rows x 24 columns]

```

```
# Display last 5 rows
```

```
df.tail()
```

	product_id		title
product_description \			
995	13698258	KREDO	Men FORTE Silver Automatic Skeleton Analogue W...
996	16896006	GUESS	Women White Patterned Dial & White Straps Anal...
997	1708108	GIORDANO	Women Rose Gold-Toned Mirrored Analogue Watch ...
998	19280114	Fastrack	Men Black Brass Dial & Black Stainless Steel S...
999	18601682	BOLDFIT	Purple Textured Yoga Mat

	rating	ratings_count	initial_price	discount	final_price
currency \					
995	3.5	11	11890	50.0	"₹11,890.00"
INR					
996	5.0	22	8900	45.0	"₹8,900.00"
INR					
997	2.8	5	7950	50.0	"₹7,950.00"
INR					
998	4.3	17	3295	20.0	"₹2,636.00"
INR					
999	3.8	161	699	31.0	"₹699.00"
INR					

	images	...	\
995	http://assets.myntassets.com/assets/images/136...	...	
996	http://assets.myntassets.com/assets/images/168...	...	
997	http://assets.myntassets.com/assets/images/170...	...	
998	http://assets.myntassets.com/assets/images/192...	...	
999	http://assets.myntassets.com/assets/images/186...	...	

	amount_of_stars
what_customers_said \	
995	{"1_star":3,"2_stars":0,"3_stars":2,"4_stars":...
NaN	
996	{"1_star":0,"2_stars":0,"3_stars":0,"4_stars":...
NaN	
997	{"1_star":2,"2_stars":1,"3_stars":0,"4_stars":...
NaN	
998	{"1_star":1,"2_stars":2,"3_stars":0,"4_stars":...
NaN	
999	{"1_star":24,"2_stars":8,"3_stars":22,"4_stars":...
NaN	

seller_name	sizes \
-------------	---------

```

995      NaN [{"size": "Onesize"}]
996      NaN [{"size": "Onesize"}]
997      NaN [{"size": "Onesize"}]
998  Tru[REDACTED]m R[REDACTED]il [{"size": "Onesize"}]
999      NaN [{"size": "Medium"}]

                                videos \
995                                ["pi-13698258"]
996  ["rw-16896006_cae700", "https://videos.myntasse...
997                                NaN
998                                NaN
999                                NaN

                                seller_information \
995                                NaN
996                                NaN
997                                NaN
998  Sho[REDACTED]o. [REDACTED]-10[REDACTED]enu[REDACTED]...
999                                NaN

                                variations \
995                                NaN
996                                NaN
997  [{}, {}, {}, {}, {}, {}, {}, {}]
998                                NaN
999  [{}, {}, {}, {}]

                                best_offer \
995                                {}
996                                {}
997                                {}
998  {"applicable_on": "Orders above Rs. 1499 (only ...
999                                {}

                                more_offers  category
995  [{"offer_name": "10% Instant Discount on Citi C...  watches
996  [{"offer_name": "10% Instant Discount on Citi C...  watches
997  [{"offer_name": "10% Instant Discount on Citi C...  watches
998  [{"offer_name": "10% Instant Discount on IndusI...  watches
999  [{"offer_name": "10% Instant Discount on Citi C...  yoga-mats

```

```
[5 rows x 24 columns]
```

```

# Display number of rows and columns i.e. the shape of the dataset
print("Shape of Dataset:", df.shape)

Shape of Dataset: (1000, 24)

# Display all column names
print(df.columns)

```

```
Index(['product_id', 'title', 'product_description', 'rating',
      'ratings_count',
      'initial_price', 'discount', 'final_price', 'currency',
      'images',
      'delivery_options', 'product_details', 'breadcrumbs',
      'product_specifications', 'amount_of_stars',
      'what_customers_said',
      'seller_name', 'sizes', 'videos', 'seller_information',
      'variations',
      'best_offer', 'more_offers', 'category'],
      dtype='str')
```

Display data types of columns

```
df.dtypes
```

```
product_id          int64
title               str
product_description str
rating              float64
ratings_count       int64
initial_price       int64
discount            float64
final_price         str
currency            str
images              str
delivery_options    str
product_details     str
breadcrumbs         str
product_specifications str
amount_of_stars     str
what_customers_said str
seller_name         str
sizes               str
videos              str
seller_information  str
variations          str
best_offer          str
more_offers         str
category            str
dtype: object
```

Display dataset information

```
df.info()
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 24 columns):
```

#	Column	Non-Null Count	Dtype
0	product_id	1000 non-null	int64

1	title	1000	non-null	str
2	product_description	1000	non-null	str
3	rating	1000	non-null	float64
4	ratings_count	1000	non-null	int64
5	initial_price	1000	non-null	int64
6	discount	879	non-null	float64
7	final_price	1000	non-null	str
8	currency	1000	non-null	str
9	images	1000	non-null	str
10	delivery_options	1000	non-null	str
11	product_details	1000	non-null	str
12	breadcrumbs	1000	non-null	str
13	product_specifications	1000	non-null	str
14	amount_of_stars	1000	non-null	str
15	what_customers_said	427	non-null	str
16	seller_name	699	non-null	str
17	sizes	1000	non-null	str
18	videos	219	non-null	str
19	seller_information	699	non-null	str
20	variations	438	non-null	str
21	best_offer	1000	non-null	str
22	more_offers	1000	non-null	str
23	category	1000	non-null	str

dtypes: float64(2), int64(3), str(19)

memory usage: 187.6 KB

Display statistical summary

df.describe()

	product_id	rating	ratings_count	initial_price
discount				
count	1.000000e+03	1000.0000	1000.000000	1000.00000
	879.000000			
mean	1.713437e+07	3.6215	76.691000	2723.24100
	53.503982			
std	3.813766e+06	1.3744	241.114263	2408.69736
	18.029201			
min	5.868460e+05	0.0000	0.000000	249.00000
	1.000000			
25%	1.511501e+07	3.7000	7.000000	1399.00000
	44.000000			
50%	1.820890e+07	4.1000	17.000000	1999.00000
	56.000000			
75%	1.902737e+07	4.3000	58.000000	3299.00000
	66.000000			
max	2.274707e+07	5.0000	4441.000000	22199.00000
	88.000000			

Handling Missing Values

In this section, missing values in the dataset are identified and handled appropriately.

The following operations are performed:

- Counting missing values in each column
- Calculating total missing values
- Filling missing numerical values using mean
- Filling missing categorical values using placeholder values
- Tracking how many values were filled

```
# Checking missing values in each column
```

```
missing_values_before = df.isnull().sum()
```

```
print("Missing Values Before Cleaning:\n")
```

```
print(missing_values_before)
```

Missing Values Before Cleaning:

product_id	0
title	0
product_description	0
rating	0
ratings_count	0
initial_price	0
discount	121
final_price	0
currency	0
images	0
delivery_options	0
product_details	0
breadcrumbs	0
product_specifications	0
amount_of_stars	0
what_customers_said	573
seller_name	301
sizes	0
videos	781
seller_information	301
variations	562
best_offer	0
more_offers	0
category	0
dtype: int64	

```
# Total missing values in dataset
```

```
total_missing_before = df.isnull().sum().sum()
```

```
print("Total Missing Values Before Cleaning:", total_missing_before)
Total Missing Values Before Cleaning: 2639
```

Dropping Columns with Excessive Missing Values

Some columns contain a very high number of missing values and are not important for this basic data cleaning assignment.

The following columns are removed:

- `videos`
- `what_customers_said`
- `variations`
- `seller_information`

These columns mainly contain multimedia or highly complex nested information.

```
# Columns to drop
columns_to_drop = [
    'videos',
    'what_customers_said',
    'variations',
    'seller_information'
]

# Dropping columns
df = df.drop(columns=columns_to_drop)

print("Dropped Columns:")
print(columns_to_drop)

Dropped Columns:
['videos', 'what_customers_said', 'variations', 'seller_information']
```

Filling Remaining Missing Values

After removing highly incomplete columns, the remaining missing values are handled.

- Numerical values are filled using mean
- Categorical values are filled using 'Unknown'

```
# Missing values before filling
discount_missing_before = df['discount'].isnull().sum()
seller_missing_before = df['seller_name'].isnull().sum()
```

```

# Filling missing numerical values
df['discount'] = df['discount'].fillna(df['discount'].mean())

# Filling missing categorical values
df['seller_name'] = df['seller_name'].fillna('Unknown')

# Checking missing values after cleaning
remaining_missing = df.isnull().sum()

print("Remaining Missing Values:\n")
print(remaining_missing)

Remaining Missing Values:
product_id          0
title               0
product_description  0
rating              0
ratings_count       0
initial_price       0
discount            0
final_price         0
currency            0
images              0
delivery_options    0
product_details     0
breadcrumbs         0
product_specifications  0
amount_of_stars     0
seller_name         0
sizes               0
best_offer          0
more_offers         0
category            0
dtype: int64

# Cleaning summary
print("Data Cleaning Summary\n")

print(f"Total Missing Values Before Cleaning: {total_missing_before}")
print(f"Columns Dropped: {len(columns_to_drop)}")
print(f"Discount Values Filled: {discount_missing_before}")
print(f"Seller Name Values Filled: {seller_missing_before}")

```

```
total_filled = discount_missing_before + seller_missing_before
print(f"Total Values Filled: {total_filled}")

remaining_total_missing = df.isnull().sum().sum()

print(f"Remaining Missing Values After Cleaning:
{remaining_total_missing}")
```

Data Cleaning Summary

```
Total Missing Values Before Cleaning: 2639
Columns Dropped: 4
Discount Values Filled: 121
Seller Name Values Filled: 301
Total Values Filled: 422
Remaining Missing Values After Cleaning: 0
```

Performing Basic Data Operations

In this section, basic data manipulation operations are performed on the dataset.

The following tasks are covered:

- Selecting specific columns
- Filtering rows based on conditions
- Displaying useful subsets of the dataset

Selecting Important Columns

Only important columns related to product information and pricing are selected for analysis.

```
# Selecting important columns
```

```
selected_columns = df[
    [
        'title',
        'rating',
        'ratings_count',
        'final_price',
        'discount',
        'category'
    ]
]
```

```
# Display selected columns
```

```
selected_columns.head()
```

```
      title  rating  ratings_count  final_price  discount
category
```

0	Lino Perros backpacks	3.8	15	"₹3,995.00"	58.0	
1	Tommy Hilfiger backpacks	4.5	67	"₹2,899.00"	35.0	
2	Lavie backpacks	4.4	226	"₹2,999.00"	65.0	
3	F Gear backpacks	4.4	1052	"₹1,675.00"	52.0	
4	MYTRIDENT robe	4.7	12	"₹2,899.00"	17.0	bath-

Filtering High Rated and Highly Discounted Products

Products with:

- rating greater than 4
- discount greater than 50%

are filtered from the dataset.

Filtering products

```
filtered_products = df[
    (df['rating'] > 4) &
    (df['discount'] > 50)
]
```

Display filtered data

```
filtered_products.head()
```

	product_id	title	
	product_description \		
2	17633752	Lavie	Aries Women Pink Mini Backpack
3	1376949	F Gear	Unisex Navy & Grey Printed Burner Backpack
5	17198778	H&M	Beige & Tan Solid Stoneware Soap Dispenser
7	18602872	My Room	Yellow & Green Floral 140 TC King Bedsheet with...
8	18602850	My Room	Ethnic Print 144 TC Super King Bedsheet with 2...

	rating	ratings_count	initial_price	discount	final_price
	currency \				
2	4.4	226	2999	65.000000	"₹2,999.00"
	INR				
3	4.4	1052	1675	52.000000	"₹1,675.00"
	INR				
5	4.5	42	1399	53.503982	"₹1,399.00"

INR					
7	4.7	10	2999	54.000000	"₹2,999.00"
INR					
8	4.5	22	2999	54.000000	"₹2,999.00"
INR					

	images	\
2	http://assets.myntassets.com/assets/images/176...	
3	http://assets.myntassets.com/assets/images/137...	
5	http://assets.myntassets.com/assets/images/171...	
7	http://assets.myntassets.com/assets/images/186...	
8	http://assets.myntassets.com/assets/images/186...	

	delivery_options	\
2	["100% Original Products","Pay on delivery mig...	
3	["100% Original Products","Pay on delivery mig...	
5	["100% Original Products","Pay on delivery mig...	
7	["100% Original Products","Pay on delivery mig...	
8	["100% Original Products","Pay on delivery mig...	

	product_details	\
2	{"description":"Pink solid backpacks Non-padd...	
3	{"description":"Navy and grey printed backpack...	
5	{"description":"Soap dispenser in glazed stone...	
7	{"description":"Set content: 1 king bedsheet w...	
8	{"description":"Dimensions: Super King bedshee...	

	breadcrumbs	\
2	[{"name":"Accessories","url":"https://www.mynt...	
3	[{"name":"Accessories","url":"https://www.mynt...	
5	[{"name":"Unisex","url":"https://www.myntra.co...	
7	[{"name":"Unisex","url":"https://www.myntra.co...	
8	[{"name":"Unisex","url":"https://www.myntra.co...	

	product_specifications	\
2	[{"specification_name":"Add-Ons","specificatio...	
3	[{"specification_name":"Compartment Closure","...	
5	[{"specification_name":"Finish","specification...	
7	[{"specification_name":"Bed Size","specificati...	
8	[{"specification_name":"Bed Size","specificati...	

	amount_of_stars	\
2	{"1_star":9,"2_stars":5,"3_stars":10,"4_stars"...	
3	{"1_star":33,"2_stars":21,"3_stars":69,"4_star...	
5	{"1_star":2,"2_stars":2,"3_stars":0,"4_stars":...	
7	{"1_star":0,"2_stars":0,"3_stars":0,"4_stars":...	
8	{"1_star":1,"2_stars":1,"3_stars":1,"4_stars":...	

	seller_name	\
2	Unknown	

```

3                                     Unknown
5  H & ████████ Hen ████████ & ████████ eta ████████
7                                     Unknown
8                                     Unknown

                                     sizes \
2                                     [{"size": "S"}]
3                                     [{"size": "Onesize"}]
5                                     [{"size": "Onesize"}]
7 [{"size": "Single"}, {"size": "Double Queen"}, {"s...
8                                     [{"size": "Super King"}]

                                     best_offer \
2                                     {}
3                                     {}
5 {"applicable_on": "Orders above Rs. 1499 (only ...
7                                     {}
8                                     {}

                                     more_offers
category
2 [{"offer_name": "10% Instant Discount on Citi C...
backpacks
3 [{"offer_name": "10% Instant Discount on Citi C...
backpacks
5 [{"offer_name": "10% Instant Discount on IndusI... bathroom-
accessories
7 [{"offer_name": "10% Instant Discount on Citi C...
bedsheets
8 [{"offer_name": "10% Instant Discount on Citi C...
bedsheets

# Number of filtered products

print("Number of High Rated & Highly Discounted Products:")

print(filtered_products.shape[0])

Number of High Rated & Highly Discounted Products:
319

```

Removing Duplicate Records

Duplicate records can affect data quality and analysis results.

In this section:

- Duplicate rows are identified
- Total duplicate records are counted
- Duplicate rows are removed from the dataset

```
# Counting duplicate rows
duplicate_count = df.duplicated().sum()
print("Total Duplicate Rows:", duplicate_count)
Total Duplicate Rows: 0
```

Removing Duplicate Rows

Duplicate rows are removed using the `drop_duplicates()` function to improve dataset quality.

Note: Here there are no duplicates in this dataset/dataframe but there could be, for those we can use following code snippet to remove duplicates.

```
# Shape before removing duplicates
shape_before = df.shape

# Removing duplicate rows
df = df.drop_duplicates()

# Shape after removing duplicates
shape_after = df.shape

print("Shape Before Removing Duplicates:", shape_before)
print("Shape After Removing Duplicates:", shape_after)
Shape Before Removing Duplicates: (1000, 20)
Shape After Removing Duplicates: (1000, 20)

# Duplicate removal summary
rows_removed = shape_before[0] - shape_after[0]
print("Duplicate Removal Summary\n")
print(f"Duplicate Rows Found: {duplicate_count}")
print(f"Rows Removed: {rows_removed}")
print(f"Final Number of Rows: {shape_after[0]}")
Duplicate Removal Summary
Duplicate Rows Found: 0
```

Rows Removed: 0
Final Number of Rows: 1000

Cleaning and Transforming Price Data

The `final_price` column contains currency symbols and commas, which makes it unsuitable for mathematical operations.

In this section:

- Currency symbols are removed
- Commas are removed
- The column is converted into numeric datatype
- Statistical analysis is performed on cleaned prices

```
# Checking datatype before cleaning
```

```
print("Datatype Before Cleaning:")
```

```
print(df['final_price'].dtype)
```

```
Datatype Before Cleaning:  
str
```

Removing Currency Symbols and Converting to Numeric Format

The `final_price` column currently contains:

- Currency symbols (₹)
- Commas (,)
- Extra quotation marks (")

These characters are removed before converting the column into float datatype.

```
# Cleaning final_price column
```

```
df['final_price'] = (  
    df['final_price']  
    .astype(str)  
    .str.replace('₹', '', regex=False)  
    .str.replace(',', '', regex=False)  
    .str.replace('"', '', regex=False)  
    .str.strip()  
)
```

```
# Converting to numeric datatype
```

```

df['final_price'] = pd.to_numeric(df['final_price'])
print("Price Column Cleaned Successfully")
Price Column Cleaned Successfully
# Checking datatype after cleaning
print("Datatype After Cleaning:")
print(df['final_price'].dtype)
Datatype After Cleaning:
float64
# Displaying cleaned price values
df['final_price'].head()
0    3995.0
1    2899.0
2    2999.0
3    1675.0
4    2899.0
Name: final_price, dtype: float64

```

Statistical Summary of Final Price

Basic statistical analysis is performed on the cleaned `final_price` column.

```

# Statistical summary
df['final_price'].describe()
count    1000.000000
mean     1706.096000
std      1783.864472
min       199.000000
25%       649.000000
50%      1099.000000
75%      2082.250000
max      17995.000000
Name: final_price, dtype: float64

```

Creating Derived Columns

The dataset does not contain a `quantity` column.

For demonstration and analysis purposes:

- A sample quantity column is created
- A new derived column `total_amount` is generated using:

```
total_amount = final_price × quantity
```

```
# Creating random quantity values between 1 and 5
```

```
np.random.seed(42)
```

```
df['quantity'] = np.random.randint(1, 6, size=len(df))
```

```
# Display quantity column
```

```
df[['final_price', 'quantity']].head()
```

	final_price	quantity
0	3995.0	4
1	2899.0	5
2	2999.0	3
3	1675.0	5
4	2899.0	5

Creating Total Amount Column

A new column named `total_amount` is created using:

```
total_amount = final_price × quantity
```

```
# Creating total_amount column
```

```
df['total_amount'] = df['final_price'] * df['quantity']
```

```
# Displaying new columns
```

```
df[['final_price', 'quantity', 'total_amount']].head()
```

	final_price	quantity	total_amount
0	3995.0	4	15980.0
1	2899.0	5	14495.0
2	2999.0	3	8997.0
3	1675.0	5	8375.0
4	2899.0	5	14495.0

```
# Statistical summary of total amount
```

```
df['total_amount'].describe()
```

count	1000.000000
mean	5027.805000

```
std      5651.825322
min      219.000000
25%     1498.500000
50%     3196.000000
75%     6420.750000
max     53996.000000
Name: total_amount, dtype: float64
```

Saving the Cleaned Dataset

After completing all preprocessing and transformation steps, the cleaned dataset is saved as a new CSV file for future analysis and usage.

```
# Saving cleaned dataset

output_file = "cleaned_dataset.csv"

df.to_csv(output_file, index=False)

print(f"Cleaned dataset saved successfully as '{output_file}'")

Cleaned dataset saved successfully as 'cleaned_dataset.csv'
```

Final Project Summary

In this project, an ecommerce dataset was explored and cleaned using Python and the Pandas library. The dataset contained product-related information such as product titles, ratings, prices, discounts, seller details, and categories.

Tasks Performed

- Imported required libraries such as:
 - Pandas
 - NumPy
- Loaded the ecommerce CSV dataset into a Pandas DataFrame using `read_csv()`.
- Explored the dataset using:
 - `head()`
 - `tail()`
 - `shape`
 - `columns`
 - `dtypes`
 - `info()`

- `describe()`
- Identified and handled missing values in multiple columns.
- Removed unnecessary columns containing sparse and complex data.
- Performed filtering and column selection operations for better analysis.
- Checked and removed duplicate records from the dataset.
- Cleaned the `final_price` column by removing currency symbols and converting it into numeric datatype.
- Created a new `quantity` column for demonstration purposes.
- Generated a derived column:
`total_amount = final_price × quantity`
- Saved the cleaned dataset as a new CSV file named:
`cleaned_dataset.csv`

Conclusion

This project helped in understanding the basics of:

- Data exploration
- Data preprocessing
- Data cleaning
- Data transformation using Pandas

After performing all cleaning and transformation steps, the dataset became more structured and ready for further analysis and visualization tasks.